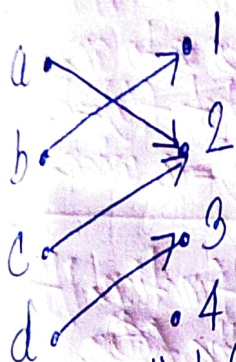
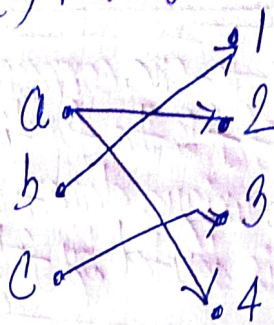


(d) Neither one-to-one nor onto



(e) Not a function



Suppose that f is a function from a set A to itself. If A is finite, then f is one-to-one if and only if it is onto. (See Ex-74). This is not necessarily the case if A is infinite (as will be shown in Sec-2.6).

Example 18: Let A be a set. The identity function on A is the function $I_A: A \rightarrow A$, where $I_A(x) = x$ for all $x \in A$. In other words, the identity function is the function that assigns each element to itself. The function I_A is one-to-one and onto, so it is a bijection. (Note that i is the Greek letter iota). $\square$

Summary

Suppose that $f: A \rightarrow B$.

To show that f is injective: Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not injective: Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective: Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not surjective: Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

~~QED~~